get representative numbers to study the characteristics of data. The characteristics of the data set is explored with some numerical measures namely measures of central tendency, measures of dispersion, measures ofskewness, and measures of kurtosis. This unit focuses on "Measure of central tendency". The measures of central tendency are also called "the averages".

In practical situations one need to have a single value to represent each variable in the whole set of data. Because, the values of the variable are not equal, however there is a general tendency of such observations to cluster around a particular level. In this situation it may be preferable to characterize each group of observations by a single value such that all other values clustered around it. That is why such measure is called the measure of central tendency of that group. A measure of central tendency is a representative value of the entire group of data. It describes the characteristic of the entire mass of data. It reduces the complexity of data and makes them amenable for the application of mathematical techniques involved in analysis and interpretation of data.

5.1 Definition of Measures of Central Tendency

Various statisticians have defined the word average differently. Some of the important definitions are given below:

"Average is an attempt to find one single figure to describe whole of figure"

– Clark and Sekkade

"Average is a value which is typical or representative of a set of data"

– Murray R. Speigal.

"The average is sometimes described as number which is typical of the whole group"

– Leabo.

It is clear from the above definitions that average is a typical value of the entire data and is a measures of central tendency.

5.2 Characteristics for a good statistical average

The following properties should be possessed by an ideal average.

- It should be well defined so that a unique answer can be obtained.
- It should be easy to understand, calculate and interpret.
- It should be based on all the observations of the data.

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